

One word at a time: is it faster?

A comparison of RSVP presentation, where words appear one at a fixed point on screen, with ordinary self-paced PDF reading.

200 median PDF, words/min	95 median RSVP, words/min	3 of 4 PDF faster
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The short answer

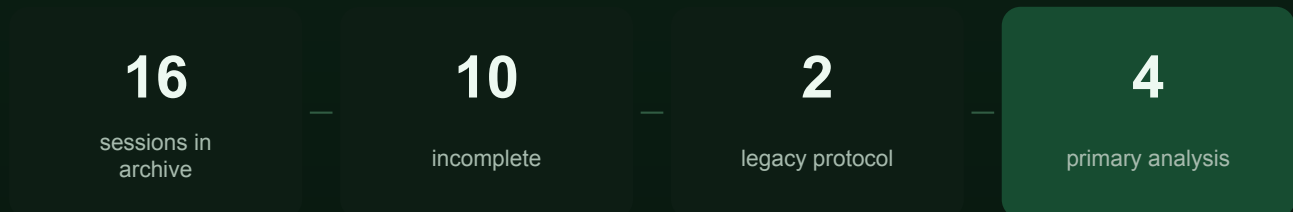
In this small pilot sample, ordinary PDF was faster for 3 of 4 participants. The median within-participant speed ratio was 1.81x in favor of PDF. But uncertainty is extremely wide, and an exact two-sided sign test does not provide statistical grounds to reject chance ($p = 0.625$).

This is not proof that PDF is universally faster. It is a transparent descriptive result from four eligible sessions, better treated as a hypothesis for the next experiment.

What was actually tested

Each participant read four Russian-language texts split into two parts. One part appeared in RSVP mode; the other opened as an ordinary PDF. Format order alternated across adjacent texts. Participants selected an RSVP pace in advance, while PDF reading was self-paced.

Data inclusion



The source directory contains 16 sessions. Ten never reached measurable reading segments. Two completed sessions used an older 12-segment protocol without a comparable comprehension rating. The primary analysis therefore includes four completed eight-segment sessions and 32 readings.

How speed was calculated

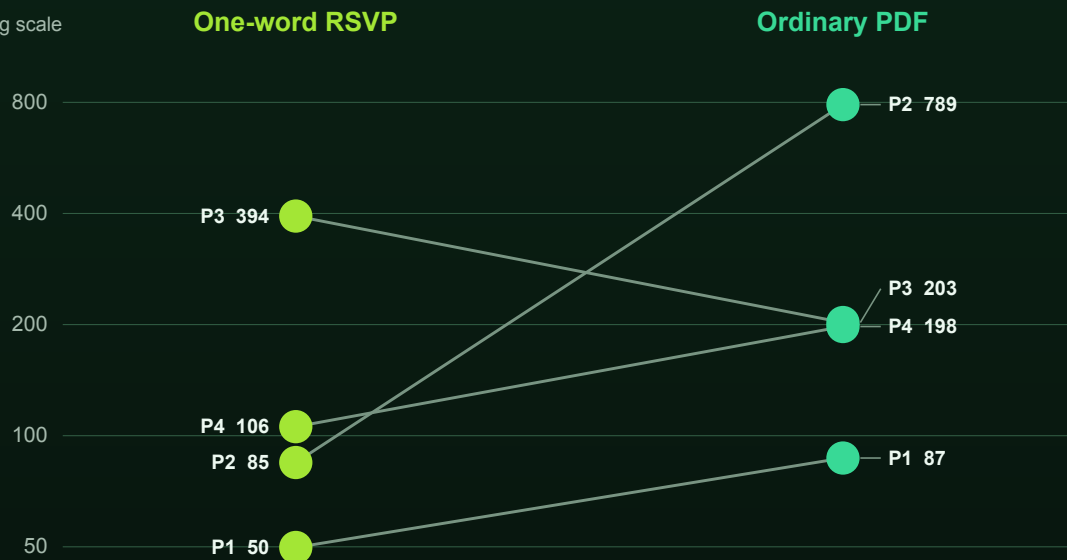
Each segment used the exact word count produced by the same regular expression as the app. Effective speed = word count / observed time x 60. Participants were compared within themselves using total words and total time in each format.

Schema recovery

The partIndex field was absent from 16 early segments. It was deterministically recovered as orderInText, following the protocol contract. Names, email addresses, source filenames, and full session identifiers are excluded from this report.

PDF was faster for three participants; RSVP for one

words/min, log scale



The chart uses a logarithmic scale to show slow and very fast observations together. P3 selected 420 words/min and completed RSVP faster than PDF. For P1, P2, and P4, effective RSVP speed was lower than ordinary reading.

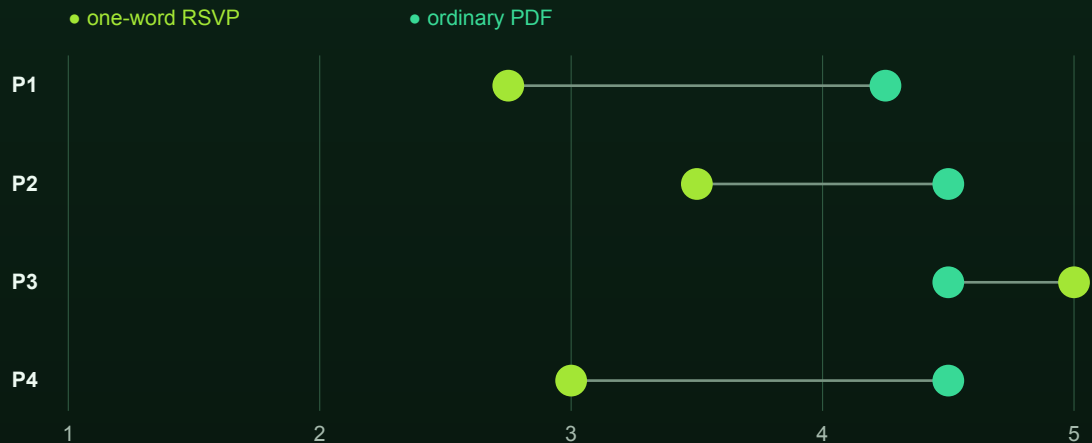
Effect estimate

Median participant speed was 200.1 words/min for PDF and 95.2 words/min for RSVP. The median PDF / RSVP ratio was 1.81x. Its bootstrap 95% interval is extremely wide: 0.51x to 9.32x. It crosses 1, making the data compatible with either an RSVP or a PDF advantage.

Data-quality check

The data contain 2 PDF segments above 1,000 words/min, both from one participant. Such values are more consistent with scanning or an early completion click than careful reading. They remain in the primary analysis because no exclusion rule was specified in advance; the conclusions therefore remain descriptive.

Self-rated comprehension also leans toward PDF



Mean participant self-rating was 4.44 of 5 for PDF and 3.56 for RSVP. The mean paired difference was +0.88 points in favor of PDF. PDF was higher for three participants and RSVP for one; two-sided sign test $p = 0.625$.

What participants wrote

Three themes recur in free-text feedback: lost context as words change, inability to linger on a difficult word, and tokenization errors. The participant with the highest selected pace explicitly said they enjoy speed reading; this was also the participant for whom RSVP was faster. This qualitative observation was not coded in advance and is not a separate test.

Anonymized results

Participant	Calibration	RSVP	PDF	PDF / RSVP	RSVP rating	PDF rating
P1	100 words/min	49.8	87.0	1.75x	2.75	4.25
P2	170 words/min	84.6	788.8	9.32x	3.50	4.50
P3	420 words/min	393.6	202.6	0.51x	5.00	4.50
P4	110 words/min	105.8	197.6	1.87x	3.00	4.50

Speed is shown in words/min; comprehension is the mean self-rating across four format segments.

The hypothesis is not supported by this dataset

Stated directly: the collected data do not show that one-word presentation is faster than ordinary text. PDF was faster in three of four eligible sessions, and those same three participants gave PDF a higher comprehension rating. The sample is too small and heterogeneous to turn that direction of effect into a general conclusion about reading formats.

The most useful pilot result is not the number 1.81x, but the apparent interaction between preferred pace, pauses, and format. RSVP may be fast for a prepared reader, while fixed pacing creates costs for readers who need to vary speed or return to context.

Primary limitations

- A very small convenience sample: $n = 4$.
- Format, text half, and order were fixed rather than randomized.
- Participants selected the RSVP pace; PDF reading was self-paced.
- RSVP pauses are included in observed time, mixing perceptual speed with interface usability.
- Comprehension was self-rated rather than tested with objective recall questions.
- Text familiarity and extraction quality may have affected the result.

How to make the next test stronger

- Randomly assign each text half to a format and counterbalance order.
- Use unfamiliar texts matched for length and difficulty.
- Record active reading time separately from pauses.
- Ask 3-5 objective literal and inferential comprehension questions after each segment.
- Predefine quality thresholds, exclusions, and the primary outcome.
- Recruit at least several dozen participants and use a paired hierarchical model.

How the result relates to prior research

RSVP research presents a mixed picture. Early experiments suggested that eliminating saccades could sharply increase measured speed. Small-screen interface studies also found RSVP advantages with comparable comprehension. More recent experiments and reviews emphasize the speed-accuracy trade-off, the loss of regressions, and poorer performance on some comprehension measures at high presentation rates.

References

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Reproducibility and privacy

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SHA-256 of the combined source JSON set: 7b45956eb070d1dc146a9f923303d9c17b2693a9a18938c91df0617d647181da